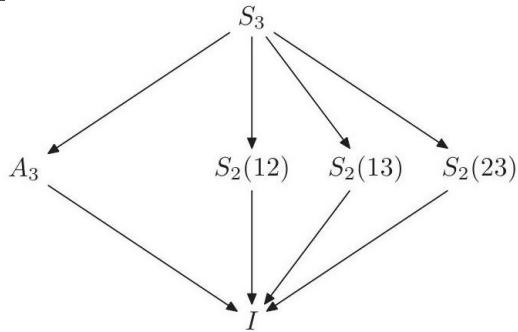
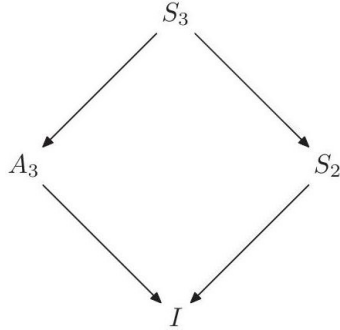
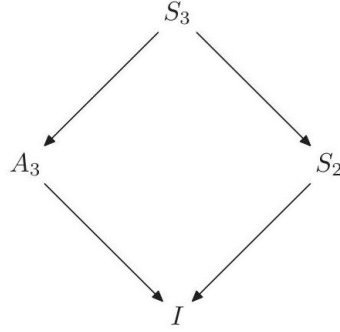
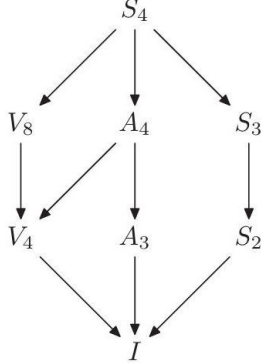
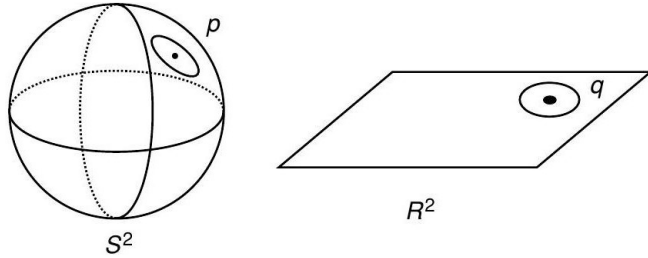
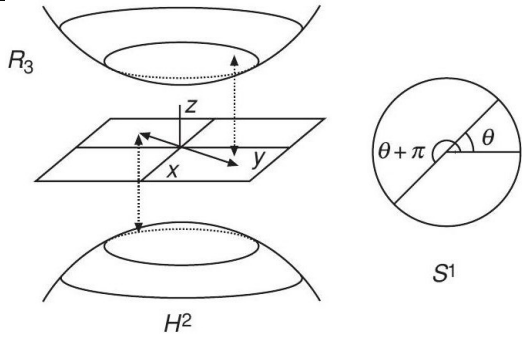
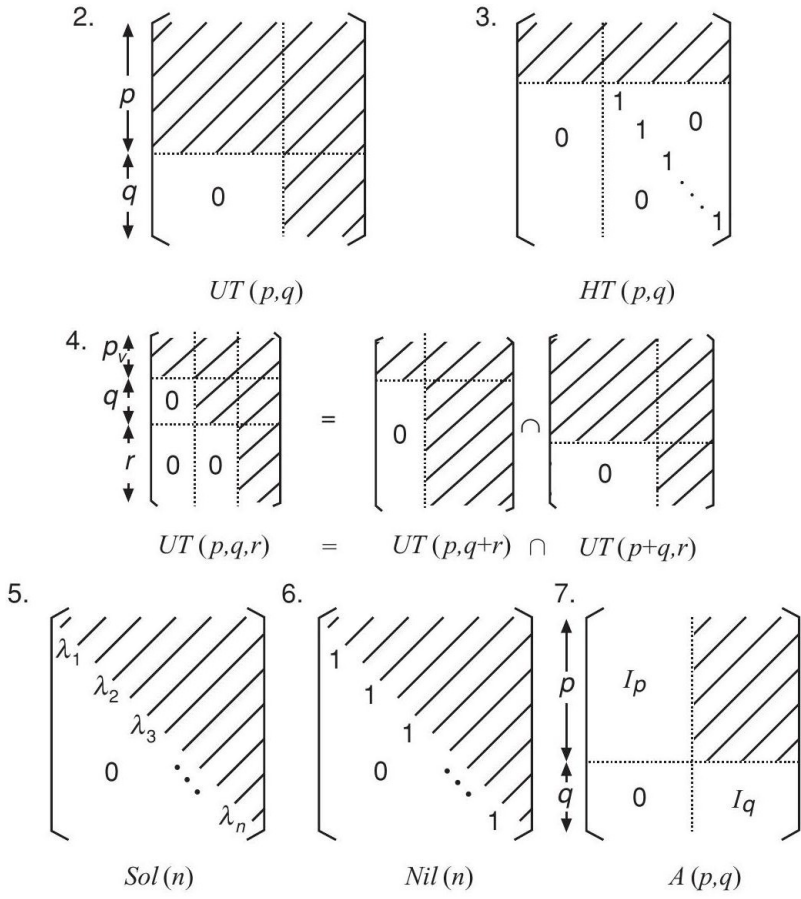
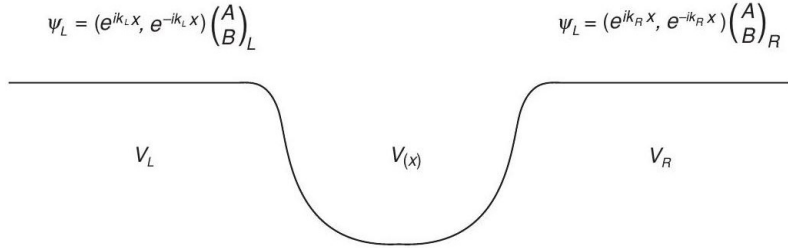


gilmore-lie-groups: image evidence

Image regions

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0001	20	—	—	—	
gilmore-lie-groups_ DIA_0002	21	—	—	—	
gilmore-lie-groups_ DIA_0003	24	—	—	—	$S_2 = \{I, (12)\}$ 
gilmore-lie-groups_ DIA_0004	26	—	—	—	
gilmore-lie-groups_ DIA_0005	29	—	—	—	

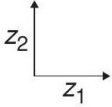
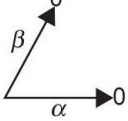
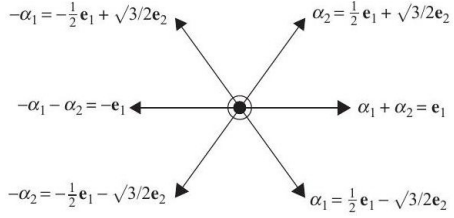
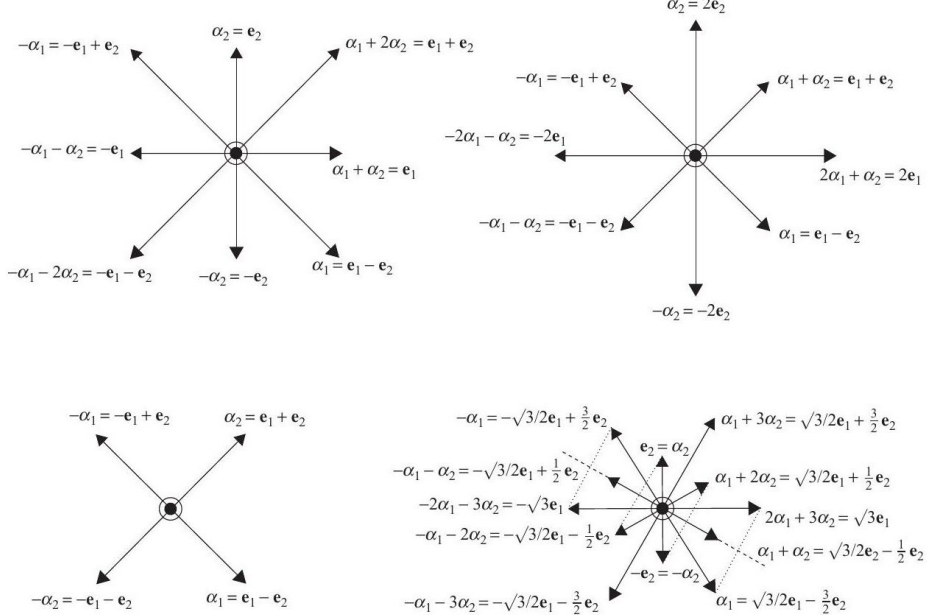
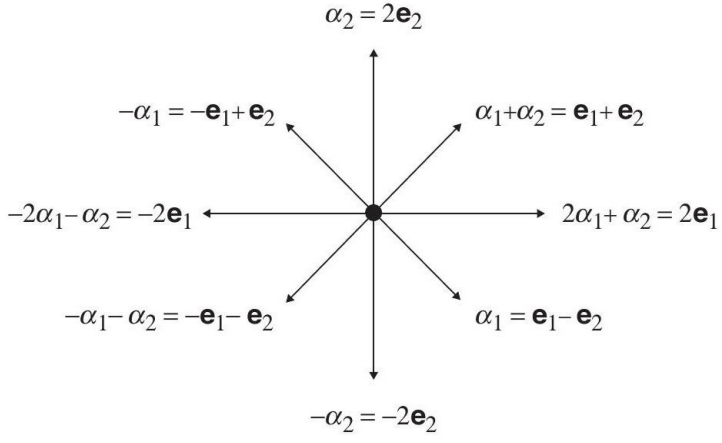
Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0006	40	—	—	—	 The diagram shows a sphere labeled S^2 with a point p on its surface. To the right, a parallelogram represents a plane labeled R^2 with a point q inside it.
gilmore-lie-groups_ DIA_0007	41	—	—	—	 The diagram shows a 3D coordinate system with axes x, y, and z. Above the xy-plane is a sphere labeled S^1 with an angle θ and $\theta + \pi$ indicated. Below the xy-plane is a hyperboloid labeled H^2. The label R_3 is also present near the top.
gilmore-lie-groups_ DIA_0008	51	—	—	—	 The diagram shows several matrix structures: 2. $UT(p,q)$: A block matrix with a shaded upper-left block of size p and a shaded lower-right block of size q, with a 0 in the bottom-left. 3. $HT(p,q)$: A block matrix with a shaded upper-left block of size p and a shaded lower-right block of size q, with a 0 in the bottom-left. 4. $UT(p,q,r) = UT(p,q+r) \cap UT(p+q,r)$: A block matrix with a shaded upper-left block of size p and a shaded lower-right block of size r, with a 0 in the bottom-left. 5. $Sol(n)$: A block matrix with a shaded upper-left block of size n and a shaded lower-right block of size n, with a 0 in the bottom-left. 6. $Nil(n)$: A block matrix with a shaded upper-left block of size n and a shaded lower-right block of size n, with a 0 in the bottom-left. 7. $A(p,q)$: A block matrix with a shaded upper-left block of size p and a shaded lower-right block of size q, with a 0 in the bottom-left.
gilmore-lie-groups_ DIA_0009	65	—	—	—	 The diagram shows a potential energy curve $V(x)$ with regions V_L, $V(x)$, and V_R. The wave function is given by $\psi_L = (e^{ik_L x}, e^{-ik_L x}) \begin{pmatrix} A \\ B \end{pmatrix}_L$ and $\psi_R = (e^{ik_R x}, e^{-ik_R x}) \begin{pmatrix} A \\ B \end{pmatrix}_R$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0010	67	—	—	—	$\begin{pmatrix} A_L \\ B_L \end{pmatrix} = T_1 \begin{pmatrix} A_M \\ B_M \end{pmatrix} \quad \begin{pmatrix} A_M \\ B_M \end{pmatrix} = T_2 \begin{pmatrix} A_R \\ B_R \end{pmatrix}$ $\begin{pmatrix} A_M \\ B_L \end{pmatrix} = S_1 \begin{pmatrix} A_L \\ B_M \end{pmatrix} \quad \begin{pmatrix} A_R \\ B_M \end{pmatrix} = S_2 \begin{pmatrix} A_M \\ B_R \end{pmatrix}$
gilmore-lie-groups_ DIA_0011	77	—	—	—	nonsemisimple reducible semisimple fully reducible simple irreducible
gilmore-lie-groups_ DIA_0012	89	—	—	—	2. $ut(p, q)$ 3. $ht(p, q)$ 4. $ut(p, q, r) = ut(p, q + r) \cap ut(p + q, r)$ 5. $sol(n)$ 6. $nil(n)$ 7. $a(p, q)$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0013	109	—	—	—	
gilmore-lie-groups_ DIA_0014	115	—	—	—	
gilmore-lie-groups_ DIA_0015	117	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0016	122	—	—	—	
gilmore-lie-groups_ DIA_0017	136	—	—	—	$ \begin{array}{ccccccc} S & & e^{ikx} & & e^{-\beta\hbar\omega a^\dagger a} & & S^{-1} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \begin{bmatrix} 1 & \alpha & \alpha\beta/2 \\ 0 & 1 & \beta \\ 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} 1 & ik/\sqrt{2} & -k^2/4 \\ 0 & 1 & ik/\sqrt{2} \\ 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 0 & 0 \\ 0 & e^{-\beta\hbar\omega} & 0 \\ 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} 1 & -\alpha & \alpha\beta/2 \\ 0 & 1 & -\beta \\ 0 & 0 & 1 \end{bmatrix} \end{array} $
gilmore-lie-groups_ DIA_0018	144	—	—	—	1 $\begin{bmatrix} & & \\ & 0 & \\ & & \end{bmatrix}$ zero 2 $\begin{bmatrix} 0 & // & \\ 0 & 0 & // \\ 0 & & \ddots \\ & & 0 \end{bmatrix}$ nil (n) 3 $\begin{bmatrix} \lambda_1 // & & \\ \lambda_2 // & // & \\ 0 & & \ddots \\ & & \lambda_n \end{bmatrix}$ sol (n) 4 $\begin{bmatrix} // & // \\ // & // \\ // & // \\ // & // \end{bmatrix}$ reducible 5 $\begin{bmatrix} // & & 0 \\ // & // & \\ 0 & & // \end{bmatrix}$ fully reducible 6 $\begin{bmatrix} // & // & // \\ // & // & // \\ // & // & // \end{bmatrix}$ irreducible
gilmore-lie-groups_ DIA_0019	144	—	—	—	$ \left[\begin{array}{c c} 0 & A \\ \hline 0 & 0 \end{array} \right] \quad \begin{array}{c} \uparrow \\ p \\ \downarrow \\ \uparrow \\ q \\ \downarrow \end{array} $
gilmore-lie-groups_ DIA_0020	148	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0021	148	—	—	—	$\left[\begin{array}{c c} * & * \\ \hline 0 & * \end{array} \right] \begin{array}{c} \uparrow \\ V_1 \\ \downarrow \\ \uparrow \\ V_2 \\ \downarrow \end{array} \Rightarrow [\text{Any}, V_2] \subset V_2$
gilmore-lie-groups_ DIA_0022	148	—	—	—	$\left[\begin{array}{c c} * & 0 \\ \hline 0 & * \end{array} \right] \begin{array}{c} \uparrow \\ V_1 \\ \downarrow \\ \uparrow \\ V_2 \\ \downarrow \end{array} \Rightarrow \begin{array}{l} [V_1, V_1] \subseteq V_1 \\ [V_2, V_2] \subseteq V_2 \\ [V_1, V_2] = 0 \end{array}$
gilmore-lie-groups_ DIA_0023	158	—	—	—	$\phi_j(a^r_s) = \frac{1}{j!(n-j)!}$
gilmore-lie-groups_ DIA_0024	158	—	—	—	$\phi_j(X^r_s) = \frac{1}{j!(n-j)!}$
gilmore-lie-groups_ DIA_0025	159	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0026	161	—	—	—	$\begin{array}{ccccc} -2 & -1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ a^2 & a & n+\frac{1}{2} & a^\dagger & a^{\dagger 2} \end{array}$ 
gilmore-lie-groups_ DIA_0027	164	—	—	—	$\begin{array}{ccccccc} \beta-m\alpha & \dots & \beta & \beta+\alpha & \dots & \beta+n\alpha \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}$ 
gilmore-lie-groups_ DIA_0028	165	—	—	—	
gilmore-lie-groups_ DIA_0029	165	—	—	—	
gilmore-lie-groups_ DIA_0030	166	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0031	176	—	—	—	
gilmore-lie-groups_ DIA_0032	177	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0033	178	—	—	—	
gilmore-lie-groups_ DIA_0034	179	—	—	—	
gilmore-lie-groups_ DIA_0035	180	—	—	—	
gilmore-lie-groups_ DIA_0036	181	—	—	—	
gilmore-lie-groups_ DIA_0037	181	—	—	—	
gilmore-lie-groups_ DIA_0038	181	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0039	183	—	—	—	$\alpha_1$$\alpha_2$$\alpha_{n-1}$$\alpha_{n-1}$ A_n $\alpha_1$$\alpha_2$$\alpha_{n-2}$$\alpha_{n-1}$$\alpha_n$ D_n $\alpha_1$$\alpha_2$$\alpha_{n-1}$$\alpha_n$ B_n $\alpha_1$$\alpha_2$$\alpha_{n-1}$$\alpha_n$ C_n $\alpha_1$$\alpha_2$ G_2 $\alpha_1$$\alpha_2$$\alpha_3$$\alpha_4$ F_4 $\alpha_1$$\alpha_2$$\alpha_3$$\alpha_4$$\alpha_5$$\alpha_6$ E_6 $\alpha_1$$\alpha_2$$\alpha_3$$\alpha_4$$\alpha_5$$\alpha_6$$\alpha_7$ E_7 $\alpha_1$$\alpha_2$$\alpha_3$$\alpha_4$$\alpha_5$$\alpha_6$$\alpha_7$ E_8
gilmore-lie-groups_ DIA_0040	184	—	—	—	$b_2^+ b_2^+$ $b_1^+ b_2^+$ $b_1^+ b_1^+$ $b_1^+ b_2$ $b_2 b_2$ $b_1 b_2$ $b_1 b_1$ $b_2^+ b_2 + \frac{1}{2}$ $b_1^+ b_1 + \frac{1}{2}$ (a) $f_1 f_2$ $f_1^{\dagger} f_2^{\dagger}$ $f_1^{\dagger} f_2$ $f_2^{\dagger} f_1 + \frac{1}{2}$ $f_2^{\dagger} f_2 + \frac{1}{2}$ (b)
gilmore-lie-groups_ DIA_0041	188	—	—	—	$\mathfrak{sl}(2;\mathbb{R})$ $\mathfrak{su}(2)$ $\mathfrak{sl}(2;\mathbb{C})$ Real Lie algebra Real Complex Complex extension Lie algebra $\frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $\frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ E_{-1} H_1 E_{+1}

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ DIA_0044	222	—	—	—	
gilmore-lie-groups_ DIA_0045	241	—	—	—	Energy eigenvalues, H atom nonrelativistic (darker), relativistic (lighter, deeper)
gilmore-lie-groups_ DIA_0046	249	—	—	—	$\begin{bmatrix} G & \\ & -\frac{1}{2} \end{bmatrix} \begin{matrix} y \\ z_1 \\ z_2 \end{matrix}$ $\begin{bmatrix} G & \\ & -1 \\ & +1 \end{bmatrix} \begin{matrix} y \\ y_{N+1} \\ y_{N+2} \end{matrix}$
gilmore-lie-groups_ DIA_0047	294	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ PIC_0001	250	—	—	—	$-N^3 \times$ Energy of hydrogen atom equal spacing suggests algebra structure
gilmore-lie-groups_ PIC_0002	305	—	—	—	
gilmore-lie-groups_ PIC_0003	308	—	—	—	